

Course: B. Tech.

Branch : IT

Semester :V

Subject Code & Name: Theory of Computation(BTITOE504A)

Max Marks: 60

Date:08/02/2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks																		
Q. 1 Solve Any Two of the following.		12																		
A) Construct a nondeterministic finite automaton (without ϵ -transitions) that accepts the language of the following regular expression: $(1+110)^*0$ Transform obtained nondeterministic finite automaton to an equivalent deterministic finite automaton that accepts the same language.	CO1	6																		
B) Explain the steps required to obtain a DFA from nondeterministic finite automaton.	CO1	6																		
C) Transform the following nondeterministic finite automaton into an equivalent deterministic finite automaton.	CO1	6																		
<table border="1"> <tr> <th></th><th>a</th><th>b</th></tr> <tr> <th>$\rightarrow p$</th><td>{p, q}</td><td>{p, t}</td></tr> <tr> <th>q</th><td>{r}</td><td>$\emptyset$</td></tr> <tr> <th>r</th><td>{s}</td><td>$\emptyset$</td></tr> <tr> <th>* s</th><td>{s}</td><td>{s}</td></tr> <tr> <th>t</th><td>$\emptyset$</td><td>{s}</td></tr> </table>		a	b	$\rightarrow p$	{p, q}	{p, t}	q	{r}	$\emptyset$	r	{s}	$\emptyset$	* s	{s}	{s}	t	$\emptyset$	{s}		
	a	b																		
$\rightarrow p$	{p, q}	{p, t}																		
q	{r}	$\emptyset$																		
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* s	{s}	{s}																		
t	$\emptyset$	{s}																		
Q.2 Solve Any Two of the following.		12																		
A) Prove that the language of balanced parenthesis strings { (), (()), ((()())), ... } is not regular.	CO2	6																		
B) Write a note on pumping lemma for regular language.	CO2	6																		
C) Construct a regular expression for the language containing strings made up of 0's and 1's such that each string contains at least two 0s and at least one 1. Also construct the ϵ -NFA for the language.	CO2	6																		
Q. 3 Solve Any Two of the following.		12																		
A) Represent the language $L = \{ 0^i 1^j 0^k \mid j > i + k \}$ using CFG.	CO3	6																		

B)	Give the steps required to convert a CFG into CNF. Express the following grammar using CNF	CO3	6
	$S \rightarrow ABA$		
	$A \rightarrow aA \mid \epsilon$		
	$B \rightarrow bB \mid \epsilon$		
C)	Write a note on pumping lemma for context-free-grammar.	CO3	6
Q.4	Solve Any Two of the following.		12
A)	Explain the equivalence between PDA and CFG.	CO4	6
B)	Design the DPDA for the language : $S \rightarrow SS \mid \{S\} \mid [S] \mid \epsilon$	CO4	6
C)	Give the formal definition of PDA and design a PDA for accepting the language of palindromes over $\{a, b\}$.	CO4	6
Q. 5	Solve Any Two of the following.		12
A)	Design a Turing Machine for accepting the language $L = \{ ss \mid s \in \{a, b\}^* \}$	CO5	6
B)	Give the formal definition of Turing machine and design the Turing Machine for the language $\{0^i 1^j \mid i = j\}$. Also give the ID conversion for the string 0011.	CO5	6
C)	Design a TM for minus or proper subtraction i.e. $m - n = \max(m-n, 0)$.	CO5	6

*** End ***